Operation Resolve & Trace

Threat Intelligence & Reconnaissance Report | Network Investigations & DNS Reconnaissance
Analyst: Chimezie Horoiheoma Lazarus (Tier-1 SOC) | Target: isc.org | Public IP: 204.152.184.88 | Private subnet: 172.31.1.50/24

1. Executive Summary

The SIEM flagged anomalous traffic and DNS queries across several hosts. I investigated the assigned domain, public IP and internal subnet using nslookup, ARIN WHOIS and nmap, following the rules of engagement for this exercise.

Area	High-level finding
Ownership	isc.org and 204.152.184.88 belong to Internet Systems Consortium, Inc. (ARIN org ISC-94-Z, net ISC-NET2, 204.152.184.0/21).
DNS	www.isc.org is a CNAME to isc.map.fastlydns.net. isc.org resolves to four IPv4 and four IPv6 addresses, all in Fastly space. The assigned IP is outside that space, in ISC's own allocation.
Exposure	Of 10 ports probed on 204.152.184.88, two are open (25 SMTP, 587 submission) and eight are filtered.
Internal subnet	172.31.1.0/24 sweep reported 64 hosts up but was only about 3% complete when captured. No port or service results yet.
Reputation	VirusTotal and AbuseIPDB results were not in the captured output. Pending.
Risk assessment	**Provisional: Low / informational.** This is my own reading, based on registry ownership and a minimal, expected mail-only exposure. It is not final until reputation checks are done.

2. DNS Record Matrix

Resolver used: 10.255.255.1 (port 53). All answers were non-authoritative.

Query	Type	Result
www.isc.org	CNAME	isc.map.fastlydns.net
isc.map.fastlydns.net	A	199.232.58.217
isc.map.fastlydns.net	AAAA	2a04:4e42:4b::729
isc.org	A	151.101.2.217 151.101.66.217 151.101.130.217 151.101.194.217
isc.org	AAAA	2a04:4e42::729 2a04:4e42:200::729 2a04:4e42:400::729 2a04:4e42:600::729
isc.org	MX	**Not captured.** Run dig isc.org MX and nslookup -type=MX isc.org.
isc.org	NS	**Not captured.** Run dig isc.org NS.
204.152.184.88	PTR	**Not captured.** Run dig -x 204.152.184.88 +short.

Analysis

• The 151.101.x.x and 199.232.x.x addresses are consistent with a CDN front end. Traffic to isc.org web content is likely served by Fastly rather than from ISC's own network (inferred from the CNAME target and address ranges, not independently verified).
• The assigned IP 204.152.184.88 does not appear in any of the DNS answers above, so it is not the web front end. Its role (likely mail, given the open ports) should be confirmed through MX and PTR records.

Task 4: dig vs nslookup for MX

Pending data. Run both commands and compare: (1) the MX hostnames and priority values should match; (2) dig prints the full response with header flags, TTLs and the answer/authority/additional sections, while nslookup prints a shorter, friendlier summary labelled non-authoritative or authoritative; (3) nslookup may omit TTLs and additional-section glue records. I have not stated results for this comparison because the output was not supplied.

3. OSINT & Geolocation Analysis

ARIN WHOIS for 204.152.184.88 (data as returned by the registry):

Field	Value
NetRange / CIDR	204.152.184.0 - 204.152.191.255 / 204.152.184.0/21

Field	Value
NetName / Handle	ISC-NET2 / NET-204-152-184-0-1
Parent / NetType	NET204 (NET-204-0-0-0-0) / Direct Allocation
Net registered / updated	1997-02-26 / 2012-04-02
Organization	Internet Systems Consortium, Inc. (ISC-94-Z)
Registered address	PO Box 360, Newmarket, NH 03857, US
Org registered / updated	2007-11-08 / 2024-01-03
Abuse contact	Internet Systems Consortium Abuse Team (ISCAT-ARIN)
NOC contact	Internet Systems Consortium NOC (ISCN-ARIN)
OriginAS	Blank in ARIN output. Confirm ASN and ISP via bgp.he.net.
Country	US (registry data; confirm with IPLocation / ip2location)

Reputation source	Status
VirusTotal (detections)	Pending: not run or not supplied
AbuseIPDB (abuse confidence score)	Pending: not run or not supplied
IPLocation / ip2location	Pending: not run or not supplied
Hurricane Electric BGP (ASN, prefixes)	Pending: not run or not supplied

Contact details for the registry organization are public, which is useful for abuse reporting if reputation checks later show malicious activity.

4. Port & Exposure Validation

4.1 Public host: 204.152.184.88

Command: `nmap -Pn -sS -p 22,25,53,80,110,143,443,587,993,995 204.152.184.88`. Host up, 0.48 s latency, scan took 8.57 s.

Port	Service	State	Interpretation
22/tcp	ssh	filtered	No response; likely firewalled
25/tcp	smtp	**open**	Mail transfer reachable from scan source
53/tcp	domain	filtered	DNS over TCP not reachable
80/tcp	http	filtered	Not exposed to this vantage point
110/tcp	pop3	filtered	Legacy mail retrieval not exposed
143/tcp	imap	filtered	Not exposed
443/tcp	https	filtered	Not exposed
587/tcp	submission	**open**	Mail submission reachable; should require STARTTLS and auth
993/tcp	imaps	filtered	Not exposed
995/tcp	pop3s	filtered	Not exposed

Limits: filtered means no reply came back. One scan, one source; no service versions (-sV) collected for this host.

4.2 Internal subnet: 172.31.1.0/24

Command: `nmap -Pn -sS -sV 172.31.1.0/24`. Nmap reported 64 hosts up and 64 undergoing SYN stealth scan. Progress stalled near 3% after about 16 minutes, with an estimated 7 to 8 hours remaining. Repeated notices said RTTVAR had grown past 2.3 seconds, which indicates slow, lossy or rate-limited responses.

Metric	Observed
Hosts up	64
Elapsed at last status	0:16:10
Scan progress	about 3.31% (peaked at 3.31%)
Estimated time remaining	about 7:45:16
Open ports / services found	None reported yet

Why it was slow: -sV adds service probing, -Pn skips discovery so every address is probed, and timing backs off when replies are delayed. A narrower first pass (top ports, -T4, --max-retries 1) then -sV on live ports would finish much faster.

5. CLI Execution Log

Terminal output as captured (long nmap progress lines trimmed).

```
$ nslookup www.isc.org
Server:         10.255.255.1
Address:        10.255.255.1#53

Non-authoritative answer:
www.isc.org     canonical name = isc.map.fastlydns.net.
Name:   isc.map.fastlydns.net
Address: 199.232.58.217
Name:   isc.map.fastlydns.net
Address: 2a04:4e42:4b::729
```

```
$ nslookup isc.org
Server:         10.255.255.1
Address:        10.255.255.1#53

Non-authoritative answer:
Name:   isc.org   Address: 151.101.2.217
Name:   isc.org   Address: 151.101.194.217
Name:   isc.org   Address: 151.101.130.217
Name:   isc.org   Address: 151.101.66.217
Name:   isc.org   Address: 2a04:4e42::729
Name:   isc.org   Address: 2a04:4e42:600::729
Name:   isc.org   Address: 2a04:4e42:400::729
Name:   isc.org   Address: 2a04:4e42:200::729
```

```
$ nmap -Pn -sS -p 22,25,53,80,110,143,443,587,993,995 204.152.184.88
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-14 11:51 -0500
Nmap scan report for 204.152.184.88
Host is up (0.48s latency).

PORT    STATE    SERVICE
22/tcp  filtered ssh
25/tcp  open     smtp
53/tcp  filtered domain
80/tcp  filtered http
110/tcp filtered pop3
143/tcp filtered imap
443/tcp filtered https
587/tcp open     submission
993/tcp filtered imaps
995/tcp filtered pop3s

Nmap done: 1 IP address (1 host up) scanned in 8.57 seconds
```

```
$ nmap -Pn -sS -sV 172.31.1.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-14 11:50 -0500
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0   (repeated)
Stats: 0:13:01 elapsed; 0 hosts completed (64 up), 64 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 2.96% done; ETC: 19:01 (6:57:59 remaining)
...
Stats: 0:16:10 elapsed; 0 hosts completed (64 up), 64 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 3.31% done; ETC: 19:51 (7:45:16 remaining)
```

6. Mitigation / SOC Recommendations

Priority	Action	Rationale
High	Complete VirusTotal, AbuseIPDB and ASN checks, then finalize the risk score and correlate with the original SIEM alerts.	Reputation is the main missing input to the risk rating.
High	Finish the internal sweep with a scoped scan (top ports, -T4, limited retries), then -sV on live ports.	64 live hosts are unassessed; a full run would take roughly 8 hours.
Medium	Verify SMTP (25) and submission (587) are expected on 204.152.184.88. In an authorized test, confirm no open relay, STARTTLS is offered, and 587 requires authentication.	Open mail ports are the only exposure found and are common abuse targets.
Medium	Capture dig MX, NS and PTR output and complete the dig vs nslookup comparison.	Required tasks not yet evidenced.
Medium	Add the domain and IP to SIEM watchlists; alert on unusual DNS query volume to isc.org and fastlydns.net names.	Supports triage of the original anomalous DNS alerts.

Note: testing covered course-assigned assets only. Pending items were not in the captured output and are not estimated; the risk rating and CDN reading are my inferences.